REPUBLIC OF CAMEROON — Peace · Work · Fatherland

« RÉUSSIR SON EXAMEN » APP — GCE SECTION

GCE A LEVEL — PURE MATHEMATICS

Practice Paper WITH FULL SOLUTIONS — 2 h

SECTION A : QUESTIONS

1. Differentiate $y = 3x^3 - 5x^2 + 2$ with respect to x.

2. Evaluate the definite integral of $(2x + 3)\,dx$ from $x = 1$ to $x = 3$.

3. Solve the equation $x^2 - 5x + 6 = 0$.

4. Write down the first three terms of the binomial expansion of $(1 + x)^5$.

5. Solve for x : $\log_2 x = 5$.

6. A geometric series has first term $a = 2$ and common ratio $r = 1/2$. Find its sum to infinity.

SECTION B : FULL SOLUTIONS

1. $dy/dx = 9x^2 - 10x$. (Differentiate term by term : $d/dx(3x^3) = 9x^2$; $d/dx(-5x^2) = -10x$; $d/dx(2) = 0$.)

2. A primitive of $2x + 3$ is $x^2 + 3x$. Then $[x^2 + 3x]$ from 1 to 3 $= (9 + 9) - (1 + 3) = 18 - 4 = 14$.

3. Factorise : $x^2 - 5x + 6 = (x - 2)(x - 3) = 0$, therefore $x = 2$ or $x = 3$.

4. $(1 + x)^5 = 1 + 5x + 10x^2 + \ldots$ (coefficients from Pascal's triangle : 1, 5, 10).

5. $\log_2 x = 5$ means $x = 2^5 = 32$.

6. For $|r| < 1$, $S_\infty = a / (1 - r) = 2 / (1 - 1/2) = 2 / (1/2) = 4$.